

研究报告：基于 Reti-Pioneer 的单调质量路由与端点感知多任务学习扩展

日期： 2026-09-14 (review-fixes)

SYNTHETIC metrics ≠ clinical AUROC

封面信息

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1. 摘要

Reti-Pioneer 发表文本描述多任务筛查，但发布训练代码为疾病独立二分类循环，质量权重固定 1/0.5/0。本仓库实现单调有界质量路由、掩码多任务 BCE、endpoint-aware cross-cohort evaluation。校准/ECE/按病种 DCA 为评估框架（非新颖性主卖点）；NB@0.10 仅示意。当前数值为 SYNTHETIC 流水线自检，不得作为临床性能。

2. 创新边界

| ID | 创新 | 状态 |

3. 方法映射

QualityAware.quality_router · ensemble=released_code · label_map.Endpoint · split.calibration_idx · masked BCE · evaluate --paper-mode · configs/ablation_e0..e4.yaml。详见 docs/paper/manuscript.md。

4. 结果（SYNTHETIC）

Arm · Eval · AUROC · AUROC_D · ECE · Cal ECE · NB@0.10 · n · Note

<figure><img

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TKJFi
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0qITp
DWJAR
FXROR
y1qPE

5. 讨论与局限

须区分 published multitask vs released independent loops; diabetes_related 不得进临床主表。待补充：真实 ODIR/BRSET 像素、GPU 特征。工程侧已补：val→test 阈值冻结、patient-level bootstrap CI、E5 门控、 λ_q 、intervention smoke。

6. 验收

见 docs/chatgpt-runs/2026-09-13-review-fixes/ACCEPTANCE.md。



fig2_ablation_bars

1.0
0.8
0.6
0.4
0.2
0.0

AUROC

